

1

In this question, the unit vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are in the directions east and north.

Distance is measured in metres and time, t , in seconds.

A radio-controlled toy car moves on a flat horizontal surface. A child is standing at the origin and controlling the car.

When $t = 0$, the displacement of the car from the origin is $\begin{pmatrix} 0 \\ -2 \end{pmatrix}$ m, and the car has velocity $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ m s⁻¹.

The acceleration of the car is constant and is $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ m s⁻².

(i) Find the velocity of the car at time t and its speed when $t = 8$.

[4]

(ii) Find the distance of the car from the child when $t = 8$.

[4]

2 The directions of the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are east and north.

The velocity of a particle, $\mathbf{v}$ m s⁻¹, at time t s is given by

$$\mathbf{v} = (16 - t^2)\mathbf{i} + (31 - 8t)\mathbf{j}.$$

Find the time at which the particle is travelling on a bearing of 045° and the speed of the particle at this time.

[6]

- 3 The unit vectors $\mathbf{i}$ and $\mathbf{j}$ shown in Fig. 2 are in the horizontal and vertically upwards directions.

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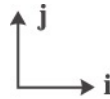


Fig. 2

Forces $\mathbf{p}$ and $\mathbf{q}$ are given, in newtons, by $\mathbf{p} = 12\mathbf{i} - 5\mathbf{j}$ and $\mathbf{q} = 16\mathbf{i} + 1.5\mathbf{j}$.

- (i) Write down the force $\mathbf{p} + \mathbf{q}$ and show that it is parallel to $8\mathbf{i} - \mathbf{j}$.

[3]

- (ii) Show that the force $3\mathbf{p} + 10\mathbf{q}$ acts in the horizontal direction.

[2]

- (iii) A particle is in equilibrium under forces $k\mathbf{p}$, $3\mathbf{q}$ and its weight $\mathbf{w}$.

Show that the value of k must be -4 and find the mass of the particle.

[3]

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- 4 The map of a large area of open land is marked in 1 km squares and a point near the middle of the area is defined to be the origin. The vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are in the directions east and north.

At time t hours the position vectors of two hikers, Ashok and Kumar, are given by:

$$\text{Ashok} \quad \mathbf{r}_A = \begin{pmatrix} -2 \\ 0 \end{pmatrix} + \begin{pmatrix} 8 \\ 1 \end{pmatrix} t,$$

$$\text{Kumar} \quad \mathbf{r}_K = \begin{pmatrix} 7t \\ 10 - 4t \end{pmatrix}.$$

- (i) Prove that the two hikers meet and give the coordinates of the point where this happens.

[4]

- (ii) Compare the speeds of the two hikers.

[3]

- 5 A particle is initially at the origin, moving with velocity $\mathbf{u}$. Its acceleration $\mathbf{a}$ is constant.

At time t its displacement from the origin is $\mathbf{r} = \begin{pmatrix} x \\ y \end{pmatrix}$, where $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \end{pmatrix} t - \begin{pmatrix} 0 \\ 4 \end{pmatrix} t^2$.

- (i) Write down $\mathbf{u}$ and $\mathbf{a}$ as column vectors.

[2]

- (ii) Find the speed of the particle when $t = 2$.

[3]

- (iii) Show that the equation of the path of the particle is $y = 3x - x^2$.

[3]

- 6 A model boat has velocity $\mathbf{v} = ((2t - 2)\mathbf{i} + (2t + 2)\mathbf{j}) \text{ m s}^{-1}$ for $t \geq 0$, where t is the time in seconds. $\mathbf{i}$ is the unit vector east and $\mathbf{j}$ is the unit vector north. When $t = 3$, the position vector of the boat is $(3\mathbf{i} + 14\mathbf{j}) \text{ m}$.

(a) Show that the boat is never instantaneously at rest. [2]

(b) Determine any times at which the boat is moving directly northwards. [2]

(c) Determine any times at which the boat is north-east of the origin. [5]

- 7 A toy car moves on a horizontal surface. Its velocity in m s^{-1} is given by

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$$\mathbf{v} = 1.5\mathbf{i} + 0.5t\mathbf{j}$$

where $\mathbf{i}$ and $\mathbf{j}$ are unit vectors east (x -direction) and north (y -direction) respectively and t is the time in seconds.

Initially the car is at the point 2 m north of the origin.

- (a) Calculate the speed of the car after 3 seconds. [2]

- (b) Find the position vector of the car after t seconds. [3]

- (c) Show that the cartesian equation of the path of the car is $y = \frac{x^2}{9} + 2$. [3]

In this question, $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are unit vectors in the x - and y -directions.

A bird is flying in the vertical plane defined by these directions.

The origin is a point on the ground.

The position vector, $\mathbf{r}$ m, of the bird at time t seconds, where $t \geq 0$, is given by

$$\mathbf{r} = \begin{pmatrix} 0 \\ 8 \end{pmatrix} + \begin{pmatrix} 2 \\ -4 \end{pmatrix}t + \begin{pmatrix} 0 \\ 1 \end{pmatrix}t^2.$$

(i) Find the velocity of the bird when $t = 2.5$.

[3]

(ii) Find the time at which the speed of the bird is 10 m s^{-1} .

[3]

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(iii) Find the times at which the bird is flying at an angle of 45° to the horizontal.

[2]

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- 9 The position vector $\mathbf{r}$ metres of a particle at time t seconds is given by

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$$\mathbf{r} = (1 + 12t - 2t^2) \mathbf{i} + (t^2 - 6t) \mathbf{j}.$$

- (a) Find an expression for the velocity of the particle at time t .

[2]

- (b) Determine whether the particle is ever stationary.

[2]

10 In this question the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

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A model boat sails from a point A with an initial velocity of $-2\mathbf{j} \text{ m s}^{-1}$. It accelerates uniformly to a velocity of $(4\mathbf{i} + 6\mathbf{j})\text{ms}^{-1}$ in 8 s.

(a) Calculate the acceleration of the boat.

[2]

(b) Find the time at which the boat is sailing due east.

[2]

4 s after leaving A, the boat is at point B with position vector $(5\mathbf{i} - 2\mathbf{j})\text{m}$.

(c) Find the position vector of A.

[4]



- 11 In this question the positive x and y directions are east and north respectively.
A model boat sails from the origin with initial velocity 3ms^{-1} due west and moves with acceleration

$$\begin{pmatrix} -0.1 \\ 0.2 \end{pmatrix} \text{ms}^{-2} \text{ for } 25 \text{ s.}$$

- (a) Show that the velocity of the boat after 25 s is $\begin{pmatrix} -5.5 \\ 5 \end{pmatrix} \text{ms}^{-1}$. [3]

- (b) Find the cartesian equation of the path of the boat. [3]

12 A goalkeeper kicks a football from ground level on a level playing field. The ball is in the air for 3.5 s.

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(a) State a modelling assumption in the standard projectile model. [1]

(b) Calculate the vertical component of the initial velocity of the ball. [2]

(c) Calculate the maximum height of the ball. [2]

(d) The ball lands 42 m from its original position. Calculate

(i) the initial speed of the ball, [3]

(ii) the angle that the initial velocity makes with the ground. [2]

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13(a) In this question, the unit vector $\mathbf{i}$ is horizontal and the unit vector $\mathbf{j}$ is vertically upwards.

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A particle of mass 0.8 kg moves under the action of its weight and two forces given by $(k\mathbf{i} + 5\mathbf{j})$ N and $(4\mathbf{i} + 3\mathbf{j})$ N. The acceleration of the particle is vertically upwards.

Write down the value of k .

[1]

(b) Initially the velocity of the particle is $(4\mathbf{i} + 7\mathbf{j}) \text{ ms}^{-1}$.

Find the velocity of the particle 10 seconds later.

[4]

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1		i	$v = u + at$	M1	May be implied by either of the next two answers but not the final answer. Evidence of use of vectors in question necessary.
		i	Velocity $\mathbf{v} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2-t \\ t \end{pmatrix}$	A1	
		i	When $t = 8$, $\mathbf{v} = \begin{pmatrix} -6 \\ 8 \end{pmatrix}$	A1	May be implied by the final answer
		i	speed $\sqrt{(-6)^2 + 8^2} = 10 \text{ m s}^{-1}$	A1	Cao but condone no units Give SC2 for 10 without working
		ii	$\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$	4	Use of correct equation with substitution. Condone omission of $\mathbf{r}_0$.
		ii		M1	Or equivalent equation
		ii	$\mathbf{r} = \begin{pmatrix} 0 \\ -2 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \end{pmatrix} \times 8 + \frac{1}{2} \times \begin{pmatrix} -1 \\ 1 \end{pmatrix} \times 8^2$	A1	Condone omission of $\mathbf{r}_0$. Follow through for their value of $\mathbf{v}$
		ii	$\mathbf{r} = \begin{pmatrix} -16 \\ 30 \end{pmatrix}$	A1	Cao but may be implied by a correct final answer.
		ii	Distance = 34 m	A1	Allow for 35.77... from $\mathbf{r} = \begin{pmatrix} -16 \\ 32 \end{pmatrix}$ and 37.57... from $\mathbf{r} = \begin{pmatrix} -16 \\ 34 \end{pmatrix}$ Examiner's Comments This question was about motion in two dimensions using column vectors. It was well answered. Such marks as were lost were usually as a result of candidates not fully answering the questions, omitting the velocity at time t and the speed in part (i) and the distance travelled in part (ii).
Total				8	
2			Equate i and j components of $\mathbf{v}$	M1	The candidate recognises that the i and j components must be equal.
			$16 - t^2 = 31 - 8t$	A1	An equation is formed.
			$t^2 - 8t + 15 = 0$		
			$(t - 3)(t - 5) = 0$		
			$t = 3 \text{ or } 5$	A1	May be implied by later working.

Question			Answer/Indicative content	Marks	Guidance
			When $t = 3$, $\mathbf{v} = 7\mathbf{i} + 7\mathbf{j}$	B1	This mark is dependent on obtaining A1 for the result $t = 3$ or 5. It is awarded if the speed for the case when $t = 5$ is not included (since $t = 5 \Rightarrow \mathbf{v} = -9\mathbf{i} - 9\mathbf{j}$ and the bearing is 225°). Note: Candidates who obtain r and equate the east and north components should be awarded SC1 for the whole question. The candidate recognises that the i and j components must be equal. This can be demonstrated during the question either by a suitable convincing diagram including 45°, or by a suitable convincing argument. Trial and error is used $t = 3$ is found by trial and error Note Candidates who obtain r and equate the east and north components should be awarded SC1 for the whole question.
			Speed when $t = 3$ is $7\sqrt{2} = 9.9 \text{ m s}^{-1}$	B1	
			The values of the i and j components must both be positive for the bearing to be 045° .	B1	
			☐ Alternative trial and error		
			The i and j components of $\mathbf{v}$ must be equal	M1	
			The i and j components of $\mathbf{v}$ must both be positive for the bearing to be 045° .	B1	
			At least one value of t is substituted	A1	
			$t = 3$	A1	
			When $t = 3$, $\mathbf{v} = 7\mathbf{i} + 7\mathbf{j}$	B1	
			Speed when $t = 3$ is $7\sqrt{2} = 9.9 \text{ m s}^{-1}$	B1	

Question			Answer/Indicative content	Marks	Guidance
					<u>Examiner's Comments</u> In this question, candidates were given the velocity of a particle using i, j notation to denote east and north, and they were asked to find when it was travelling on a compass bearing of 045° and its speed at that time. This involved equating the components of $\mathbf{v}$; this gave a quadratic equation, leading to two possible times. Candidates then had to recognise that at one of these times the bearing was 225° not 45°. Many candidates obtained full marks on this question. A few made the mistake of trying to work with position vector instead of the velocity. A common mistake was to fail to eliminate the 225° case. A small number of candidates set out to answer this question using a trial and error method and some credit was given for this.
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
3		i	$\mathbf{p} + \mathbf{q} = 28\mathbf{i} - 3.5\mathbf{j}$	B1	Or equivalent. k may be implied by going straight to 3.5
		i	$28\mathbf{i} - 3.5\mathbf{j} = k(8\mathbf{i} - \mathbf{j})$	M1	
		i	$k = 3.5$ (So they are parallel)	A1	
			Alternative		Comparing the ratio of the components in each of the two vectors is sufficient, using any consistent sign convention. The angle does not need to be worked out, nor does $\tan$ have to be seen.
		i	$\mathbf{p} + \mathbf{q} = 28\mathbf{i} - 3.5\mathbf{j}$ $\mathbf{p} + \mathbf{q}: \tan \theta = \frac{-3.5}{28} \Rightarrow \theta = -7.13^\circ$	B1	
		i	$8\mathbf{i} - \mathbf{j}: \tan \theta = \frac{-1}{8} \Rightarrow \theta = -7.13^\circ$	M1	
		i	So they are parallel	A1	Both ratios the same and correct
		ii	$3\mathbf{p} + 10\mathbf{q} = (36 + 160)\mathbf{i} + (-15 + 15)\mathbf{j}$		Or equivalent explanation. May be shown on a diagram
		ii	$= 196\mathbf{i}$	B1	
		ii	Zero $\mathbf{j}$ component so horizontal	B1	
		iii	The horizontal component must be zero		Substituting $k = -4$ and showing $\mathbf{i}$ component is zero is acceptable Award for 24.5 seen Award for 2.5 seen. FT from their weight.
		iii	So $12k + 3 \times 16 = 0 \Rightarrow k = -4$	B1	Substituting $k = -4$ and showing $\mathbf{i}$ component is zero is acceptable
		iii	$\mathbf{w} = -24.5\mathbf{j}$	B1	Award for 24.5 seen

Question			Answer/Indicative content	Marks	Guidance
		iii	$mg = 24.5 \Rightarrow m = 2.5$ The mass is 2.5 kg	B1	Award for 2.5 seen. FT from their weight. Examiner's Comments This question was about vectors defined using i, j notation. It was well answered with many candidates obtaining full marks. Most of the errors that did occur were in part (iii) where candidates were required to interpret the vectors as forces on a particle in equilibrium, with its weight now introduced. Some did not interpret the equilibrium conditions sufficiently rigorously, failing to recognise that both the horizontal and vertical components of the resultant had to be zero. The question ended with a request for the mass of the particle and some candidates confused this with its weight.
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
4		i	Either $-2 + 8t = 7t$ Or $t = 10 - 4t$	M1	Forming an equation for t . Accept vector equation for this mark. May be implied by a statement that $t = 2$.
		i	$\Rightarrow t = 2$	A1	
		i	Substituting $t = 2$ in both expressions	B1	oe, eg showing $t = 2$ satisfies both equations or a vector equation.
		i	They meet at (14, 2)	B1	Accept $\begin{pmatrix} 14 \\ 2 \end{pmatrix}$ Examiner's Comments Candidates were asked to prove the two hikers meet and this involved showing their position vectors are the same at some time ($t = 2$). Many lost a mark by not showing that this was true for both components when the position vectors were equated.
		ii	Ashok's speed is $\sqrt{8^2 + 1^2} = \sqrt{65}$	B1	
		ii	Kumar's speed is $\sqrt{7^2 + (-4)^2} = \sqrt{65} \text{ km h}^{-1}$	B1	
		ii	They both walk at the same speed	B1	CAO from correct speeds SC1 for finding both velocities correctly but neither speed Examiner's Comments Candidates were asked to compare the speeds of the two hikers and most got this right. A few compared their velocities. Others did not know how to obtain the velocities from the given expressions in terms of t for the position vectors.
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
5		i	$\mathbf{u} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$	B1	Examiner's Comments In this question the position of a particle at time t was given as a column vector. In part (i) candidates were asked to write down $\mathbf{u}$ and $\mathbf{a}$ as column vectors. Most were successful in this but a common mistake was to give $\mathbf{v}$ instead of $\mathbf{u}$. Answer $\begin{pmatrix} 2 \\ 6 \end{pmatrix}, \begin{pmatrix} 0 \\ -8 \end{pmatrix}$
		i	$\mathbf{a} = \begin{pmatrix} 0 \\ -8 \end{pmatrix}$	B1	
		ii	$\mathbf{v} = \mathbf{u} + \mathbf{a}t$		Examiner's Comments In the next part candidates were asked to find the speed at a certain time and this was well answered with many recovering from errors in part (i). Follow through was allowed for the values of $\mathbf{u}$ and $\mathbf{a}$ that they found in part (i). Common mistakes were sign errors and not distinguishing between speed and velocity. Answer $\begin{pmatrix} 2 \\ -10 \end{pmatrix}, 10.2 \text{ m s}^{-2}$
		ii	$t = 2 \Rightarrow \mathbf{v} = \begin{pmatrix} 2 \\ 6 \end{pmatrix} + \begin{pmatrix} 0 \\ -8 \end{pmatrix} \times 2$	M1	
		ii	$= \begin{pmatrix} 2 \\ -10 \end{pmatrix}$	A1	
		ii	speed = $\sqrt{2^2 + (-10)^2} = 10.2 \text{ ms}^{-1}$ (to 3sf)	B1	
		iii	$x = ut \square x = 2t \square t = \frac{x}{2}$	M1	This mark may also be obtained for substituting x for $2t$ in the expression for y.
		iii	$y = 6t - 4t^2$	B1	

Question			Answer/Indicative content	Marks	Guidance
		iii	$y = 6 \times \frac{x}{2} - 4 \times \left(\frac{x}{2}\right)^2 = 3x - x^2$	A1	Examiner's Comments In the final part candidates were asked to show that the position vector at time t led to a given cartesian equation for the path of the particle. This was answered confidently and almost entirely successfully.
		iii	Alternative $x = 2t$		
		iii	Substitute for x in given answer	M1	
		iii	$y = 3x - x^2 \quad \square \quad y = 6t - 4t^2$	A1	
		iii	This is the given expression for y	B1	
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
6		a	Require both components zero at the same <i>time</i> i component zero only when $t = 1$ and j component only when $t = -1$ so there are no such times	M1(AO3.1b) A1(AO2.4) [2]	May be implied but must be clear Or say j component ≥ 2 since $t \geq 0$
		b	This requires use of the velocity vector Travelling due north means that the i component is zero and the j component +ve So we need $2t - 2 = 0$ for i component, giving $t = 1$. This gives j component $4 > 0$ so yes at $t = 1$.	M1(AO3.3) A1(AO2.4) [2]	Recognise velocity vector required Must test j component
		c	This requires use of the position vector either $\mathbf{r} = \int \mathbf{v} dt$ so $\mathbf{r} = \int ((2t - 2)\mathbf{i} + (2t + 2)\mathbf{j}) dt = (t^2 - 2t + C)\mathbf{i} + (t^2 + 2t + D)\mathbf{j}$ $\mathbf{r} = 3\mathbf{i} + 14\mathbf{j}$ when $t = 3$ so $C = 0$ and $D = -1$ so $\mathbf{r} = (t^2 - 2t)\mathbf{i} + (t^2 + 2t - 1)\mathbf{j}$ Or $\mathbf{a} = 2\mathbf{i} + 2\mathbf{j}$ when $t = 3$ $\mathbf{v} = 4\mathbf{i} + 8\mathbf{j}$, $\mathbf{r} = (4\mathbf{i} + 8\mathbf{j})(t - 3) + \frac{1}{2}(2\mathbf{i} + 2\mathbf{j})(t - 3)^2 + 3\mathbf{i} + 14\mathbf{j}$ and so $\mathbf{r} = (t^2 - 2t)\mathbf{i} + (t^2 + 2t - 1)\mathbf{j}$ the boat is NE of O when the i and j components are equal and +ve we require $t^2 - 2t = t^2 + 2t - 1$ so $t = 0.25$ this gives components of -0.4375 so no.	M1(AO3.1b) M1(AO1.1) A1(AO1.1) M1(AO1.1) A1(AO1.1) M1(AO3.2b) A1(AO2.1) [5]	Recognise position vector required May use + C instead Must find a but may omit $3\mathbf{i} + 14\mathbf{j}$ Award even if +ve not mentioned Must be complete argument
			Total	9	

Question			Answer/Indicative content	Marks	Guidance
7		a	When $t = 3$, $ v = 1.5i + 1.5j $ $= \sqrt{1.5^2 + 1.5^2}$ Speed is $\frac{3}{2}\sqrt{2} = 2.12 \text{ m s}^{-1}$	M1(AO 1.1b) A1(AO 1.1b) [3]	Substitute for t and attempt modulus
		b	$\mathbf{r} = \int \mathbf{v} \, dt = 1.5t\mathbf{i} + 0.5 \times \frac{t^2}{2} \mathbf{j} + \mathbf{c} = 1.5t\mathbf{i} + 0.25t^2\mathbf{j} + \mathbf{c}$ $t = 0$ and $\mathbf{r} = 2\mathbf{j} \Rightarrow \mathbf{c} = 2\mathbf{j}$ $\mathbf{r} = 1.5t\mathbf{i} + (0.25t^2 + 2)\mathbf{j}$	M1(AO 1.1a) A1(AO 1.1b) A1(AO 2.1) [3]	Attempt to integrate both terms Allow without + c Terms need not be collected
		c	$x = 1.5t$ and $y = 0.25t^2 + 2$ $y = 0.25\left(\frac{x}{1.5}\right)^2 + 2$ $y = \frac{x^2}{9} + 2$ AG	M1(AO 2.1) M1(AO 2.1) A1(AO 1.1b) [3]	Allow for either equation Eliminate t Must be fully justified
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
8		i	Differentiating $\mathbf{r}$ $\mathbf{v} = \begin{pmatrix} 2 \\ -4 \end{pmatrix} + \begin{pmatrix} 0 \\ 2 \end{pmatrix} t$ $\mathbf{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$	M1 A1 B1 [3]	Attempt at differentiation must be seen Apply ISW for speed = $\sqrt{5}$ providing $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ is seen. Examiner's Comments This question was about vectors, using the context of the flight of a bird. The position vector of the bird was given in term of the time. In part (i) candidates were asked about the velocity of the bird and this was well answered using vectors.

Question			Answer/Indicative content	Marks	Guidance
		ii	$\sqrt{2^2 + (-4 + 2t)^2} = 10$ $t^2 - 4t - 20 = 0$ $t = \frac{4 \pm \sqrt{4^2 - 4 \times 1 \times -20}}{2} (= 6.898... \text{ or } -2.898...)$ $t = 6.9 \text{ (or } -2.9) \text{ (to 2 sf)}$	M1 M1 A1 [3]	Attempt at formulation of the given information using their vector $\mathbf{v}$ from part (i). Must involve both components. e.g. $-4 + 2t = \sqrt{96}$ Accept drawing triangle of velocities Attempted solution of an equation for t. Dependent on previous M mark Allow FT from their vector expression for $\mathbf{v}$ in part (i). Else CAO. Condone not giving the negative value of t as well as the correct value. Dependent on both M marks. Examiner's Comments In part (ii), the time was to be found at which the bird had a given speed. This involved using a vector expression to form a scalar equation. Many candidates did not know how to go about this. Others obtained the right answer but their explanations were not always the most elegant. However there were very good answers written by some candidates.

Question			Answer/Indicative content	Marks	Guidance
		iii	Either $2 = -4 + 2t \Rightarrow t = 3$ Or $-2 = -4 + 2t \Rightarrow t = 1$	B1 B1 [2]	FT from their vector expression for v in part (i). FT from their vector expression for v in part (i). Examiner's Comments In part (iii), candidates were asked to find the times when the bird was flying at an angle of 45° to the horizontal. Correct answers to this were somewhat uncommon. One common mistake was to equate the components of the position vector rather than the velocity; most of those who did use the velocity considered only the case when the bird was flying above the horizon and not when it was flying below it.
			Total	8	

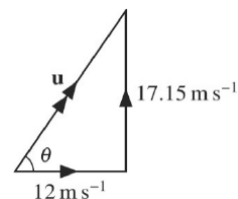
Question			Answer/Indicative content	Marks	Guidance
9		a	$\mathbf{v} = \frac{d\mathbf{r}}{dt} = (12 - 2 \times 2t)\mathbf{i} + (2t - 6)\mathbf{j}$	M1 (AO 1.1a) A1 (AO 2.5) [2]	Attempt to differentiate at least one coefficient Must use vector notation Examiner's Comments Most candidates used vector notation accurately and successfully differentiated to obtain a correct expression for velocity.
		b	When $t = 3$ both components of velocity are zero, so the particle is stationary at $t = 3$.	M1 (AO 3.1a) E1 (AO 2.2a) [2]	Equating at least one component of their vector velocity to zero Must be argued from two zero components Examiner's Comments This was typically well answered with most candidates realising the requirement for both components to be zero at the same time.  Misconception It is not sufficient to equate the components and solve to find $t = 3$. From this starting point, candidates would need to check that the components were zero to achieve the method mark.
			Total	4	

Question			Answer/Indicative content	Marks	Guidance	
10		a	$4\mathbf{i} + 6\mathbf{j} = -2\mathbf{j} + 8\mathbf{a}$ $\mathbf{a} = 0.5\mathbf{i} + \mathbf{j} \text{ m s}^{-2}$	M1 (AO 1.1a) A1 (AO 2.5) [2]	For use of vector equation $\mathbf{v} = \mathbf{u} + \mathbf{a}t$, or alternatively for finding both acceleration components separately Must be vector form; do not allow final answer as separate components	Do not allow for the magnitude of the acceleration unless final answer is labelled as such
		b	$\mathbf{v} = -2\mathbf{j} + (0.5\mathbf{i} + \mathbf{j})t$ is directed east when $-2 + t = 0$ $t = 2$, so boat sails east at time 2 s	M1 (AO 3.1b) A1 (AO 3.2a) [2]	For equating j component of velocity to zero, giving equation for t cao	
		c	Displacement in 4 s is $(-2\mathbf{j}) \times 4 + \frac{1}{2}(0.5\mathbf{i} + \mathbf{j}) \times 4^2$ $= 4\mathbf{i} (+ 0\mathbf{j})$ Position at time $t = 4$ gives $\mathbf{r}_A + 4\mathbf{i} = 5\mathbf{i} - 2\mathbf{j}$ $\mathbf{r}_A = \mathbf{i} - 2\mathbf{j}$	M1 (AO 3.1b) A1 (AO 1.1) M1 (AO 3.1b) A1 (AO 1.1) [4]	For any use of $\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$ with $t = 4$ and their $\mathbf{a}$ even if no clarity about displacement/ position For correct use of position vectors Must be in vector form	Equation may include initial position term $\mathbf{r}_A$ May be earned earlier, if original equation is $\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$
			Total	8		

Question			Answer/Indicative content	Marks	Guidance	
11		a	Initial velocity is $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$ $\mathbf{v} = \mathbf{u} + \mathbf{a}t = \begin{pmatrix} -3 \\ 0 \end{pmatrix} + \begin{pmatrix} -0.1 \\ 0.2 \end{pmatrix} 25$ $= \begin{pmatrix} -5.5 \\ 5 \end{pmatrix}$	B1 (AO 2.1) M1 (AO 2.1) A1 (AO 2.5) [3]	For correct vector seen FT their initial velocity as long as it is a vector AG; must be in vector form	Allow full credit for an answer where the two components are considered separately only if the final answer is given as a vector
		b	$\mathbf{r} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2 = \begin{pmatrix} -3 \\ 0 \end{pmatrix}t + \frac{1}{2}\begin{pmatrix} -0.1 \\ 0.2 \end{pmatrix}t^2$ $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -3t - 0.05t^2 \\ 0.1t^2 \end{pmatrix}$ Substitute $t^2 = 10y$ into equation for x $x = -3\sqrt{10y} - 0.5y$	M1 (AO 3.1a) M1 (AO 3.1a) A1 (AO 1.1b) [3]	FT their $\mathbf{u}$ Attempt to eliminate t; FT their $\mathbf{r}$	Allow full credit for both components considered separately
			Total	6		

Question			Answer/Indicative content	Marks	Guidance
12		a	Resistance to the motion is modelled as being negligible	B1 (AO 3.3) [1]	Or similar Or any correct assumption e.g. motion is in a vertical plane; football is modelled as a particle; acceleration is constant...
		b	Vertical motion: $0 = 3.5u - \frac{1}{2} \times 9.8 \times 3.5^2$ $u = 17.15$, so velocity component is 17.2 m s^{-1} (to 3sf)	M1 (AO 3.1a) A1 (AO 1.1b) [2]	Use of <i>suvat</i> equation(s) to find <i>u</i> . using either $s = 0$, $g = -9.8$ and $t = 3.5$ or $v = 0$, $g = -9.8$ and $t = 1.75$
		c	$s = 17.15 \times 1.75 - \frac{1}{2} \times 9.8 \times 1.75^2$ Maximum height is 15.0 m	M1 (AO 1.1a) A1 (AO 1.1b) [2]	Use of <i>suvat</i> equation(s) to find <i>s</i> , using any of $u =$ (their) 17.15, $v = 0$, $t = 1.75$, $g = -9.8$ Allow for 15 or better

Question			Answer/Indicative content	Marks	Guidance
		d	(i) Horizontal velocity component is $\frac{42}{3.5} = 12$ Initial speed $= \mathbf{u} = \sqrt{17.15^2 + 12^2}$ $= 20.9 \text{ m s}^{-1}$ (ii) Angle θ with horizontal is given by $\tan \theta = \frac{17.15}{12}$ Angle with horizontal is 55.0°	B1 (AO 3.1a) M1 (AO 1.1a) A1 (AO 1.1b) [3] M1 (AO 1.1a) A1 (AO 1.1b) [2]	Use of Pythagoras FT their components Allow reciprocal for M mark only FT their components; allow 55° or better
			Total	10	



Question			Answer/Indicative content	Marks	Guidance
13	a		Component in i direction zero $\Rightarrow k = -4$	B1 (AO 3.3) [1]	Allow $(-4\mathbf{i} + 5\mathbf{j})$ seen instead of k Do not allow for $k = -4\mathbf{i}$ or similar. Examiner's Comments Many candidates produced quite a bit of working for a 'Write down...' question, but many had the correct value. The mark was not given where k was given as a vector.

Question		Answer/Indicative content	Marks	Guidance	
	b	Weight = $-0.8g\mathbf{j}$ N2L $5\mathbf{j} + 3\mathbf{j} - 0.8g\mathbf{j} = 0.8\mathbf{a}$ $[\Rightarrow \mathbf{a} = 0.2\mathbf{j}]$ $\mathbf{v} = (4\mathbf{i} + 7\mathbf{j}) + 0.2\mathbf{j} \times 10$ velocity is $(4\mathbf{i} + 9\mathbf{j}) \text{ m s}^{-1}$ Alternative solution Acceleration is vertical so consider only vertical motion $5 + 3 - 0.8g = 0.8a$ $[\Rightarrow a = 0.2]$ $v = u + at = 7 + 0.2 \times 10 = 9$ So the velocity is $(4\mathbf{i} + 9\mathbf{j}) \text{ m s}^{-1}$	B1 (AO 1.2) M1 (AO 1.1a) M1 (AO 1.1a) A1 (AO 2.5) [4] M1 B1 M1 A1 [4]	Allow if seen Condone missing weight Using their $\mathbf{a}$ in a vector <i>suvat</i> equation(s) Must be in correct vector form Applying N2L in 1-dimension Condone missing weight Including the weight (consistent sign convention) Using <i>suvat</i> equation(s) in the vertical direction Must be in vector form Examiner's Comments There was considerable confusion about whether to use vectors or scalars and some candidates had equations with a mixture. More common was the omission of the weight in the N2L equation, resulting in an incorrect acceleration and eventually a very large vertical component of the velocity. Again, there was a lot of premature rounding of intermediate calculations, but there was some very efficient work from many candidates. Some realised that the fact there was a zero-horizontal component of force meant that they could work in a vertical direction until the end of the question, but the final answer had to be a vector.	Accept fully correct column vector Accept fully correct column vector
		Total	5		